

Paper Writing Assignment

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Summarize the Key Points: What are the main objectives of the paper?

The main objective of the paper is to discuss how data science analytics can further lead to data-driven smart decision making in real world scenarios. This includes detailing various types of advanced analytics methods which could result in applications making smarter decisions in different scenarios. The author gives an overview of the various machine learning models and their various real-world applications. The author also provides ten potential real-world domains with examples to see the improvement of using machine learning-based analytics.

● **TGDS Model:** What is the TGDS (Theory-Guided Data Science) model, and what is its significance in the field of data science?

TGDS is the paradigm that tries to combine previous scientific knowledge with the analytics of data science for more accurate models. The TGDS model is significant because data analytics don't always result in accurate or applicable conclusions in real world scenarios. There are times where models are able to accurately find relationships among its training and test sets, but don't generalize well. We are essentially using our accumulated scientific knowledge as a way of guiding the learned patterns of data science models to be generalizable.

● **Impact on Future Advancements:** Reflect on how the paper's focus on real-world application domains of data science could impact future advancements in the field. Provide specific examples mentioned in the paper.

Other papers cited by the article are focused on very specific domains and how data science may be used in said domains. This paper gives an overview of data science methods and models that could be applied in a broader scope to various real-world applications. I believe that this paper is a great overview of data science and the focus on real-world applications makes it easier for a reader to understand why data science analytics are so important. An example provided is the use of machine learning to predict infectious outbreaks, minimize medical costs and improve overall quality of life. Another example is related to online retailers, like Amazon, where they could use machine learning to predict out-of-stock situations or optimize warehouse logistics.

● **Types of Data:** Differentiate between the four types of data mentioned: structured, unstructured, semi-structured, and metadata. Provide examples of each type.

Structured data is data that follows a standard format such as names, dates, addresses, credit card information, and more. Unstructured data has no pre-defined organization such as emails, audio files, videos, etc. Semi-structured data is a mix of both structured and unstructured where it has some organization, this includes HTML, CML, and JSON documents. Metadata is just information describing data meaning author, file type, and file size.

● **Analytics in Decision-Making:** How can organizations use a combination of descriptive, diagnostic, predictive, and prescriptive analytics to make more informed business decisions? Provide a hypothetical example where all four types of analytics are utilized.

Organizations can use a combination of each advanced analytic to figure out what problem happened in the past, the root of the problem, what may happen in the future, and how they might combat the issue. In general, these analytics help give companies a broader picture of the possible problems they may need to fix, either in the present or in the future.

An example that uses all four analytics may be something in the retail industry. Let's say that a business sells hand wipes and there is a new virus slowly being spread more and more in the community. The company can use descriptive analytics to look at previous sales history of hand wipes when supply was high or low. They can then use diagnostic analytics to see the correlation between viral outbreaks and a lack of supply in hand wipes. With predictive analytics the company could notice that due to the increasing spread of the new virus there will be a large wave of customers purchasing hand wipes. Now with prescriptive analytics the company see the best method of increasing supply of hand wipes while decreasing supply of less likely bought items to maximize profits and keep up with demand.

● **Real-World Applications:** Identify two compelling real-world applications from the "Real-World Application Domains" section. Explain what makes them intriguing and how you would leverage machine learning to tackle their challenges. Which specific models or methods would you apply and why?

One compelling real-world application is cybersecurity. I find this application intriguing because we are in a very online world where personal information like bank accounts or passwords are enticing for hackers. The issue is that companies usually store this information so it's up to them to use machine learning to better detect possible threats. I would use machine learning to better understand areas that are weaker to attacks like DDoS or observe recent trends of strange, potentially malicious, online traffic in company hosted servers. One specific method I may apply would be anomaly detection. I could use it to single out strange user logs or devices in the network to potentially prevent a cybersecurity attack on the company's servers.

Another real-world application I'm intrigued by is behavioral data science. Things like user application preferences, browsing history, and purchase history are all incredibly important for companies to understand to keep up with competition. It's just interesting how valuable user information is now. I would likely use machine learning to detect anomalies in a user's activity. If the user's browsing history and purchase history has a dramatic shift then it's possible the user's credentials were stolen. In order to protect the user's information like saved credit cards, their account could be locked. In this case, using a method like log analysis seems to be best. We would view the user's activity on the network logs or server logs to notice any dramatic changes to their account activity. Machine learning would mainly be helpful in picking out anomalous activity from large sets of log entries.

B Real World Case Study

A real world case study of data privacy violation involved the online counseling service BetterHelp. Despite promising to keep customer health data, including information about mental health issues, BetterHelp still sold this data to third parties like Facebook and Snapchat. When applying for counseling, customers have to fill in a questionnaire asking for sensitive mental health information, including experience with depression or suicidal thoughts, as well as medication history. Customers must also provide name, email, birth date, and other personal information. BetterHelp was proposed to pay \$7.8 million by the FTC for selling their user's information to advertisers.

“FTC to Ban Betterhelp from Revealing Consumers’ Data, Including Sensitive Mental Health Information, to Facebook and Others for Targeted Advertising.” *Federal Trade Commission*, 2 Mar. 2023,

www.ftc.gov/news-events/news/press-releases/2023/03/ftc-ban-betterhelp-revealing-consumers-data-including-sensitive-mental-health-information-facebook.

- What ethical principle(s) were violated?

Various ethical principles were violated including breach of privacy, consent, and misuse of data. BetterHelp did not receive any of their users’ consent before selling their information and did not place any limits on how the third parties could use said data.

- What should have been done differently to avoid the violation?

To avoid this violation, BetterHelp should have asked for its users’ permission before disclosing their health data to advertisers. The only reason BetterHelp did not is because the service knew the majority of users would not wish for that. Better yet, BetterHelp should have never tried to sell its users’ health data in the first place.

- How can this case guide future responsible data science practices?

This case is important because it shows what not to do during the data collection process in the data science pipeline. When collecting data it is important to clearly outline what the information will be used for and be transparent about the individual’s privacy. If we should disclose the individual’s information then it is imperative to receive their consent before doing so. Overall, data privacy is one of the most important parts of data science and we must make sure that the individual feels safe in providing information for our studies.